

## MARK SCHEME for the October/November 2007 question paper

### 9709 MATHEMATICS

9709/02

Paper 2, maximum raw mark 50

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- CIE will not enter into discussions or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the October/November 2007 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.



## **Mark Scheme Notes**

Marks are of the following three types:

- M     Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.
  - A     Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).
  - B     Mark for a correct result or statement independent of method marks.
- When a part of a question has two or more "method" steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep\*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
  - The symbol  $\checkmark$  implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously "correct" answers or results obtained from incorrect working.
  - Note:     B2 or A2 means that the candidate can earn 2 or 0.  
              B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking  $g$  equal to 9.8 or 9.81 instead of 10.



The following abbreviations may be used in a mark scheme or used on the scripts:

|     |                                                                                                                                                                                         |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AEF | Any Equivalent Form (of answer is equally acceptable)                                                                                                                                   |
| AG  | Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)                                                     |
| BOD | Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)                                                                                                  |
| CAO | Correct Answer Only (emphasising that no "follow through" from a previous error is allowed)                                                                                             |
| CWO | Correct Working Only - often written by a 'fortuitous' answer                                                                                                                           |
| ISW | Ignore Subsequent Working                                                                                                                                                               |
| MR  | Misread                                                                                                                                                                                 |
| PA  | Premature Approximation (resulting in basically correct work that is insufficiently accurate)                                                                                           |
| SOS | See Other Solution (the candidate makes a better attempt at the same question)                                                                                                          |
| SR  | Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance) |

### **Penalties**

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MR -1 | A penalty of MR -1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become "follow through $\sqrt{}$ " marks. MR is not applied when the candidate misreads his own figures - this is regarded as an error in accuracy. An MR-2 penalty may be applied in particular cases if agreed at the coordination meeting. |
| PA -1 | This is deducted from A or B marks in the case of premature approximation. The PA -1 penalty is usually discussed at the meeting.                                                                                                                                                                                                                                                                                                                                     |



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|       |                                                                                                                           |     |     |
|-------|---------------------------------------------------------------------------------------------------------------------------|-----|-----|
| 1     | State indefinite integral of the form $k \ln(2x + 1)$ , where $k = \frac{1}{2}$ , 1 or 2                                  | M1  |     |
|       | State correct integral $\frac{1}{2} \ln(2x + 1)$                                                                          | A1  |     |
|       | Use limits correctly, allow use of limits $x = 4$ and $x = 1$ in an incorrect form                                        | M1  |     |
|       | Obtain given answer                                                                                                       | A1  | [4] |
| 2 (i) | Use the iterative formula correctly at least once                                                                         | M1  |     |
|       | Obtain final answer 2.29                                                                                                  | A1  |     |
|       | Show sufficient iterations to justify its accuracy to 2 d.p. (must be working to 4 d.p.)<br>– 3 iterations are sufficient | B1  | [3] |
| (ii)  | State equation $x = \frac{2}{3}x + \frac{4}{x^2}$ , or equivalent                                                         | B1  |     |
|       | Derive the exact answer $\alpha$ (or $x$ ) $= \sqrt[3]{12}$ , or equivalent                                               | B1  | [2] |
| 3 (i) | Obtain critical values 4 and 6                                                                                            | B1  |     |
|       | State answer $4 < y < 6$                                                                                                  | B1  | [2] |
| (ii)  | Use correct method for solving an equation of the form $3^x = a$ , where $a > 0$                                          | M1  |     |
|       | Obtain one critical value, i.e. either 1.26 or 1.63                                                                       | A1  |     |
|       | State answer $1.26 < x < 1.63$                                                                                            | A1  | [3] |
| 4     | State derivative $2 - \sec^2 x$ , or equivalent                                                                           | B1  |     |
|       | Equate derivative to zero and solve for $x$                                                                               | M1  |     |
|       | Obtain $x = \frac{1}{4}\pi$ , or $0.785$ ( $\pm 45^\circ$ gains A1)                                                       | A1  |     |
|       | Obtain $x = -\frac{1}{4}\pi$ , (allow negative of first solution)                                                         | A1✓ |     |
|       | Obtain corresponding $y$ -values $\frac{1}{2}\pi - 1$ and $-\frac{1}{2}\pi + 1$ , $\pm 0.571$                             | A1  | [5] |
| 5 (i) | Substitute $x = -2$ and equate to zero                                                                                    | M1  |     |
|       | Obtain answer $a = 3$                                                                                                     | A1  | [2] |
| (ii)  | At any stage state that $x = -2$ is a solution                                                                            | B1  |     |
|       | EITHER: Attempt division by $x + 2$ and reach a partial quotient of $3x^2 + kx$                                           | M1  |     |
|       | Obtain quadratic factor $3x^2 + 2x - 1$                                                                                   | A1  |     |
|       | Obtain solutions $x = -1$ and $x = \frac{1}{3}$                                                                           | A1  |     |
|       | OR: Obtain solution $x = -1$ by trial or inspection                                                                       | B1  |     |
|       | Obtain solution $x = \frac{1}{3}$ similarly                                                                               | B2  | [4] |

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- 6 (i) State answer  $R = 17$ , allow  $\sqrt{289}$  B1  
 Use trig formula to find  $\alpha$  M1  
 Obtain  $\alpha = 61.93^\circ$ , (1.08 radians) A1 [3]
- (ii) Carry out evaluation of  $\sin^{-1}(14/17) \approx 55.44^\circ$ , or equivalent M1  
 Obtain answer  $117.4^\circ$ , (2.06 radians) A1  
 Carry out correct method for second answer M1  
 Obtain answer  $186.5^\circ$  and no others in the range (3.255 radians) A1✓ [4]  
 [Ignore answers outside the given range.]
- 7 (i) Expand and use  $\sin 2A$  formula M1  
 Use  $\cos 2A$  formula at least once M1  
 Obtain any correct expression in terms of  $\cos 2x$  and  $\sin 2x$  only – can be implied A1  
 Obtain given answer correctly A1 [4]
- (ii) State indefinite integral  $5x - 2\sin 2x - \frac{3}{2}\cos 2x$  B2  
 [Award B1 if one error in one term]  
 Substitute limits correctly – must be correct limits M1  
 Obtain answer  $\frac{1}{4}(5\pi - 2)$ , or exact simplified equivalent A1 [4]
- 8 (i) Differentiate using product or quotient rule M1  
 Obtain derivative in any correct form A1  
 Equate derivative to zero and solve for  $x$  M1  
 Obtain answer  $x = 2$  correctly, with no other solution A1 [4]
- (ii) Find the gradient of the curve when  $x = 1$ , must be simplified, allow 0.368 B1  
 Form the equation of the tangent when  $x = 1$  M1  
 Show that it passes through the origin A1 [3]
- (iii) State or imply correct ordinates 0.36787..., 0.54134..., 0.44808... B1  
 Use correct formula, or equivalent, correctly with  $h = 1$  and three ordinates M1  
 Obtain answer 0.95 with no errors seen A1 [3]